PHYS 143 – Mechanics and Thermal Physics

Introduction: Physical Quantities – Notes

Hope Donglo, Office 24, Department of Physics, University of Ghana

These notes go deeper than the introduction lecture. They follow Young and Freedman, Chapter 1, and they are not needed for the first problem sets. Where a short argument goes beyond the textbook's own text, it is marked as an *extension*.

1 Base units and prefixes in full

The three base units used from Chapter 1

Quantity	Unit	Present standard
Time	second (s)	the time required for $9\,192\,631\,770$ cycles of the microwave radiation that makes the outermost electron of a cesium-133 atom reverse its spin direction; adopted in 1967
Length	metre (m)	the distance that light travels in vacuum in $1/299\,792\,458$ second; the speed of light is defined to be exactly $299\,792\,458\,\mathrm{m/s}$; adopted in November 1983
Mass	kilogram (kg)	since 2018 based on Planck's constant, defined as $h = 6.62607015 \times 10^{-34}\,\mathrm{kg \cdot m^2/s}$; the gram is 0.001 kilogram

Earlier standards were less precise: the second was a fraction of the mean solar day until 1967, the metre was tied to the wavelength of orange-red light from krypton-86 in 1960, and the kilogram was a metal cylinder kept in France until 2018. A good standard must not change and must be reproducible by observers in different places. Temperature (kelvin) and amount of substance (mole) are introduced with the thermal physics chapters.

Derived units met in this course

Quantity	Unit	In base units
Speed	$\mathrm{m/s}$	length divided by time
Force	newton (N)	$\mathrm{kg \cdot m/s^2}$
Energy	joule (J)	$\mathrm{kg \cdot m^2/s^2}$
Power	watt (W)	$\mathrm{J/s = kg \cdot m^2/s^3}$

Prefix table

The prefix names are formed by adding the prefix to the name of the base unit; for mass the prefix is added to the gram. Young prints the standard list in Appendix A; the table below is the standard SI list, so check it against your copy.

Prefix	Symbol	Factor	Prefix	Symbol	Factor
pico	p	10^{-12}	kilo	k	10^{3}
nano	n	10^{-9}	mega	M	10^{6}
micro	μ	10^{-6}	giga	G	10^{9}
milli	m	10^{-3}	tera	T	10^{12}
centi	c	10^{-2}			

Young's examples for the common ones: $1\,\mathrm{nm}$ is a few times the size of the largest atom; $1\,\mu\mathrm{m}$ the size of some bacteria; $1\,\mathrm{mm}$ the diameter of the point of a ballpoint pen; $1\,\mathrm{cm}$ the diameter of your little finger; $1\,\mathrm{km}$ the distance covered in a ten-minute walk; $1\,\mathrm{ns}$ the time for light to travel $0.3\,\mathrm{m}$.

2 Why multiplying by a unit ratio works

Rule: unit multipliers

If two quantities are the same physical quantity expressed in different units, their ratio equals 1. Multiplying by that ratio, or by its reciprocal, changes the units but not the physical quantity.

When we write $1\,\mathrm{min} = 60\,\mathrm{s}$ we do not mean that the number 1 equals the number 60. We mean that $1\,\mathrm{min}$ represents the same time interval as $60\,\mathrm{s}$. Divide both sides by $60\,\mathrm{s}$:

$$\frac{1\,\mathrm{min}}{60\,\mathrm{s}} = 1, \qquad \text{and likewise} \qquad \frac{60\,\mathrm{s}}{1\,\mathrm{min}} = 1.$$

A quantity multiplied by 1 is the same quantity, so

$$3\,\mathrm{min} = (3\,\mathrm{min})\left(\frac{60\,\mathrm{s}}{1\,\mathrm{min}}\right) = 180\,\mathrm{s}.$$

Units are treated like algebraic symbols under multiplication and division, so the min cancels. If the wrong ratio is used, the unwanted unit stays, as in $(3\,\mathrm{min})(1\,\mathrm{min}/60\,\mathrm{s}) = \frac{1}{20}\,\mathrm{min}^2/\mathrm{s}$, which is not a time. This is the built-in check.

Products of units: powers of a factor

If the units of a quantity are a product of simpler units, such as $\mathrm{m}^3 = \mathrm{m} \times \mathrm{m} \times \mathrm{m}$, use a product of unit multipliers. Starting from $1\,\mathrm{m} = 10^2\,\mathrm{cm}$,

$$1\,\mathrm{m}^3 = (1\,\mathrm{m})^3 = (10^2\,\mathrm{cm})^3 = 10^6\,\mathrm{cm}^3.$$

The number is cubed together with the unit. In the same way an area factor is squared: $1\,\mathrm{m}^2 = 10^4\,\mathrm{cm}^2$.

The rule between km/h and m/s

$$1\,\mathrm{km/h} = \frac{1000\,\mathrm{m}}{3600\,\mathrm{s}} = \frac{1}{3.6}\,\mathrm{m/s}.$$

Hence a speed in $\mathrm{m/s}$ is its value in $\mathrm{km/h}$ divided by 3.6, and a speed in $\mathrm{km/h}$ is its value in $\mathrm{m/s}$ times 3.6. For example $20\,\mathrm{m/s} = 72\,\mathrm{km/h}$ and $90\,\mathrm{km/h} = 25\,\mathrm{m/s}$.

S1. Converting a density

IDENTIFY: express the density of iron, $7.87\,\mathrm{g/cm^3}$, in $\mathrm{kg/m^3}$.
SET UP: $1\,\mathrm{g} = 10^{-3}\,\mathrm{kg}$ and $1\,\mathrm{cm^3} = 10^{-6}\,\mathrm{m^3}$.
EXECUTE: $\rho = 7.87\,\dfrac{\mathrm{g}}{\mathrm{cm^3}} \times \dfrac{10^{-3}\,\mathrm{kg}}{1\,\mathrm{g}} \times \dfrac{1\,\mathrm{cm^3}}{10^{-6}\,\mathrm{m^3}} = \boxed{7.87 \times 10^3\,\mathrm{kg/m^3}}$.
EVALUATE: the factor is $10^{-3}/10^{-6} = 10^3$, so $1\,\mathrm{g/cm^3} = 10^3\,\mathrm{kg/m^3}$. Water ($1\,\mathrm{g/cm^3}$) is $1000\,\mathrm{kg/m^3}$, and iron is a little under eight times denser.

3 Significant figures in more detail

Rule: significant figures

Multiplication or division: the result can have no more significant figures than the factor with the fewest. Addition or subtraction: the result has as many digits to the right of the decimal point as the term with the fewest. Round the final answer; keep extra digits during the calculation.

Counting

Non-zero digits always count. Zeros between non-zero digits count. Zeros to the left of the first non-zero digit do not count; they only place the decimal point, so $0.25\,\mathrm{km}$ has two significant figures and $0.250\,\mathrm{km}$ has three. In scientific notation every digit shown counts: 4.00×10^8 has three, the two trailing zeros being significant. A number like $384\,000\,000\,\mathrm{m}$ does not show how many zeros are significant, so we write $3.84 \times 10^8\,\mathrm{m}$.

Extension: why the two rules differ

Suppose a length $12.4\,\mathrm{cm}$ is uncertain by about $0.05\,\mathrm{cm}$ and a width $5.1\,\mathrm{cm}$ by about $0.05\,\mathrm{cm}$. Their fractional uncertainties are about $0.05/12.4 = 0.4\%$ and $0.05/5.1 = 1.0\%$. For a product the fractional uncertainties add roughly, giving about 1.4% of $63.24\,\mathrm{cm^2}$, which is $0.9\,\mathrm{cm^2}$. So the area is about $63 \pm 1\,\mathrm{cm^2}$: two significant figures, as the rule says. For a sum the *absolute* uncertainties add, which is why the decimal places decide. In $123.62 + 8.9$ the uncertainties are about 0.01 and 0.1, so the sum is uncertain by about 0.1: 132.5, not 132.52.

S2. Multiplying with exact and inexact numbers

IDENTIFY: the circumference C and the diameter d of a circle are measured as $424\,\mathrm{mm}$ and $135\,\mathrm{mm}$. Find $\pi = C/d$.
SET UP: each measurement has three significant figures, so the quotient has three.
EXECUTE: $\dfrac{424\,\mathrm{mm}}{135\,\mathrm{mm}} = 3.1407\ldots \Rightarrow \boxed{3.14}$.
EVALUATE: within three significant figures this agrees with the true value 3.141592654. A calculator answer with ten digits would overstate the precision.

S3. Rounding, not truncating

IDENTIFY: the ratio of $525\,\mathrm{m}$ to $311\,\mathrm{m}$ to three significant figures.
EXECUTE: $\dfrac{525}{311} = 1.68810\ldots$. The digit after the third figure is 8, so we round up: $\boxed{1.69}$, not 1.68.

EVALUATE: truncating would give 1.68, which is outside the rounded value by one unit in the last place.

Exact numbers

An integer or a fraction in an equation is treated as having no uncertainty. In $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$ the coefficient 2 and the exponents 2 are exactly 2: think of them as having an infinite number of significant figures.

4 Uncertainty notation

Definition: measurement uncertainty

Measurement uncertainty is an estimate of the range within which the true value is expected to lie. It depends on the measurement technique. The *error* is a different idea: it is the difference between the measured value and the true value, which we usually do not know. The uncertainty is our estimate of how large that difference might be.

Form	Meaning
$56.47 \pm 0.02\,\mathrm{mm}$	the true value is likely within 56.45 to $56.49\,\mathrm{mm}$
$1.6454(21)$	shorthand for 1.6454 ± 0.0021; the digits in brackets are the uncertainty in the final digits
Fractional uncertainty	$\dfrac{0.02\,\mathrm{mm}}{56.47\,\mathrm{mm}} \approx 0.0004$
Percent uncertainty	$0.0004 \times 100\% \approx 0.04\%$
$47\ \Omega \pm 10\%$	differs from $47\ \Omega$ by no more than 10%, about $5\ \Omega$; between 42 and $52\ \Omega$

When an explicit uncertainty is not given, the significant figures indicate, by convention, the resolution and precision of the measurement; they are not a rigorous uncertainty, which the instrument, the procedure and repeated readings determine: $2.91\,\mathrm{mm}$ means the first two digits are correct and the third is uncertain by about $0.01\,\mathrm{mm}$, while $137\,\mathrm{km}$ is uncertain by about $1\,\mathrm{km}$.

Accuracy versus precision

Accuracy is how close a value is likely to be to the true value. Precision describes how closely repeated measurements agree with one another; it is often reflected by the number of meaningful digits reported, but a value can have many digits and still be imprecise. A cheap digital watch showing $10{:}35{:}17$ that repeats its readings consistently is very precise but, if it runs several minutes slow, not accurate. A grandfather clock with no second hand can be accurate without being very precise. A high-quality measurement is both.

Extension: repeated measurements

Repeating a measurement and looking at the spread of the values shows the precision and gives an estimate of the uncertainty. For the times 2.1, 2.3, 2.2, $2.2\,\mathrm{s}$ the mean is $2.2\,\mathrm{s}$ and no value is more

than $0.1\,\mathrm{s}$ from it, so as an illustrative estimate from these four readings we report $(2.2 \pm 0.1)\,\mathrm{s}$. The maximum deviation is not a universal definition of uncertainty and systematic effects are not covered; a fuller analysis depends on the method. The standard deviation is the full statistical measure and is treated in PHYS 105.

5 Estimates and orders of magnitude

An order-of-magnitude estimate aims at the approximate scale of a quantity, usually the nearest power of ten, and is often good only to within a factor of a few; it is still useful. Make the best guesses you can for the data; do not look up a lot of data.

S4. Seconds in a year

Predict: about 3×10^7.
IDENTIFY: number of seconds in one year.
SET UP: $1\,\mathrm{year} = 365.25\,\mathrm{days}$, $1\,\mathrm{day} = 24\,\mathrm{h}$, $1\,\mathrm{h} = 3600\,\mathrm{s}$.
EXECUTE: $365.25 \times 24 \times 3600\,\mathrm{s} = 3.156 \times 10^7\,\mathrm{s}$.
EVALUATE: close to the prediction, and a useful number to remember: about $\pi \times 10^7\,\mathrm{s}$.

S5. Water in a swimming pool

Predict: hundreds of thousands of kilograms.
IDENTIFY: the mass of water in a pool about $25\,\mathrm{m}$ long, $10\,\mathrm{m}$ wide and $2\,\mathrm{m}$ deep.
SET UP: $m = \rho V$ with $\rho = 1000\,\mathrm{kg/m^3}$ for water.
EXECUTE: $V = 25 \times 10 \times 2\,\mathrm{m^3} = 500\,\mathrm{m^3}$, so $m = (1000\,\mathrm{kg/m^3})(500\,\mathrm{m^3}) = 5 \times 10^5\,\mathrm{kg}$, which is 500 tonnes.
EVALUATE: consistent with the prediction. $500\,\mathrm{m^3}$ is also 5×10^5 litres. The guessed dimensions may be wrong by a factor of two, so quote the order of magnitude, $10^5\,\mathrm{kg}$.

Answers to the worked examples

S1: $7.87 \times 10^3\,\mathrm{kg/m^3}$. S2: 3.14. S3: 1.69. S4: $3.16 \times 10^7\,\mathrm{s}$. S5: about $5 \times 10^5\,\mathrm{kg}$.